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Effects of attentional instructions on the behavioral and neural mechanisms of speech auditory feedback control

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Abstract

The present study investigated how instructions for paying attention to auditory feedback may affect speech error detection and sensorimotor control. Electroencephalography (EEG) and speech signals were recorded from 21 neurologically intact adult subjects while they produced the speech vowel sound /a/ and received randomized ±100 cents pitch-shift alterations in their real-time auditory feedback. Subjects were instructed to pay attention to their auditory feedback and press a button to indicate whether they detected a pitch-shift stimulus during trials. Data for this group was compared with 22 matched subjects who completed the same speech task under altered auditory feedback condition without attentional instructions. Results revealed a significantly smaller magnitude of speech compensations in the attentional-instruction *vs.* no-instruction group and a positive linear association between the magnitude of compensations and P2 event-related potential (ERP) amplitudes. In addition, we found that the amplitude of P2 ERP component was significantly larger in the attentional-instruction *vs.* no-instruction group. Source localization analysis showed that this effect was accounted for by significantly stronger neural activities in the right hemisphere insula, precentral gyrus, postcentral gyrus, transverse temporal gyrus, and superior temporal gyrus in the attentional-instruction group. These findings suggest that attentional instructions may enhance speech auditory feedback error detection, and subsequently improve sensorimotor control via generating more stable speech outputs (i.e., smaller compensations) in

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CRediT authorship contribution statement

Yilun Zhang: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Kimaya Sarmukadam:** Writing – review & editing, Supervision, Data curation. **Yuan Wang:** Writing – review & editing, Formal analysis. **Roozbeh Behroozmand:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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response to pitch-shift alterations. Our data are informative for advancing theoretical models and motivating targeted interventions with a focus on the role of attentional instructions for improving treatment outcomes in patients with motor speech disorders.

Keywords

Speech motor control; Auditory feedback; Attentional instructions; Sensorimotor integration; Event-related potential

1. Introduction

1.1. The role of auditory feedback in speech control

The unique ability to produce speech is supported by sensorimotor mechanisms that enable humans to monitor, detect, and correct for their production errors for effective verbal communication (Hickok et al., 2011). Auditory feedback provides an important source of sensory input that is used to guide fluent speech (Guenther, 2016). The critical involvement of the auditory system in speech error detection and correction is supported by evidence from previous studies showing that the vocal motor system produces a compensatory response that changes speech fundamental frequency (f_o) in the opposite direction of perceived pitch-shift alterations in the online auditory feedback (Burnett et al., 1998; JL, 1981; Larson, 1998). Speech compensation responses are also produced by changing the position of articulatory muscles to oppose online alterations in the formant frequencies of auditory feedback (Cai et al., 2011). Under delayed auditory feedback (DAF) conditions, speech production is disrupted by the reduced rate, increased intensity, increased f_o, and exaggerated (e.g., stutter-like) repetitions that occur to overcome the effect of feedback delays (Fairbanks, 1954; Yates, 1963). In addition, according to the Lombard effect, speech loudness increases in the presence of background noise (Lane and Tranel, 1971), and the amplification of auditory feedback leads to compensatory increases in speech loudness outputs (Chang-Yit et al., 1975).

According to the contemporary models of speech, the aforementioned behavioral effects in response to altered auditory feedback (AAF) are indicative of reciprocal interactions between audio-vocal brain networks implicated in the integration of sensory and motor processes during speech (Guenther, 2016; Hickok, 2012; Hickok et al., 2011, 2023; Hickok & Poeppel, 2000, 2007; Houde and Chang, 2015; Houde & Nagarajan, 2011). In this context, the activation of motor commands and their efference copy representations during speaking results in the generation of sensory predictions that modulate feedback processing in the auditory system via an internal forward model. The presence of a sensory prediction mechanism enables selective suppression of neural responses of the auditory cortex to normal speech feedback in comparison to listening to the playback of self-produced speech sounds (Behroozmand and Larson, 2011; Flinker et al., 2010; Heinks-Maldonado et al., 2005). This effect, known as speaking-induced suppression (SIS), results from the central cancellation of neural representations associated with predicted auditory feedback derived from the efference copy of intended speech motor commands compared with the actual sensory consequences of produced speech. By selectively attenuating the neural response

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to self-generated feedback, the brain can distinguish it from external auditory stimuli, facilitating accurate perception and processing of own speech. Thus, sensory predictions play a crucial role in differentiating self-generated speech from external sounds, ensuring efficient and accurate speech production and perception. In addition, evidence demonstrating neural enhancement of responses to auditory feedback alterations (i.e., errors) during speaking vs. listening tasks suggests that a predictive mechanism is crucial for error detection and subsequent corrective motor reactions (Behroozmand et al., 2009; Chang et al., 2013; Greenlee et al., 2013). Speaking-induced enhancement of neural responses to altered feedback signifies increased sensitivity to deviations between predicted and actual sensory representations. This heightened neural responsiveness enables the brain to efficiently detect and recognize errors in speech production (Behroozmand et al., 2009; Behroozmand and Larson, 2011; Chang et al., 2013; Flinker et al., 2010; Greenlee et al., 2013; Heinks-Maldonado et al., 2005). Accordingly, the discrepancy between predicted and actual feedback plays a crucial role in driving compensatory behavior for speech motor control. When the brain detects a mismatch between the expected and actual feedback, it triggers corrective (i.e., compensatory) responses to reduce the disparity and improve speech accuracy. This entails real-time adjustments in motor commands that drive vocal or articulatory movements during speech production. The interplay between feedback error detection and the subsequent compensatory behavior lies in the pivotal role of comparing the internally predicted sensory representations with actual feedback for monitoring and adjustment processes that drive corrective actions to maintain the desired speech output.

1.2. Attentional modulation of speech sensorimotor processing

Despite significant advancements in our understanding of theoretical and neuro-computational models of speech, the effects of higher-level cognitive processes (e.g., attention) on these mechanisms remains insufficiently investigated. Recent studies have revealed the instructional effects of divided attention on sensorimotor mechanisms of speech auditory feedback control during the processing of irrelevant visual stimuli (Liu et al., 2015, 2018; Scheerer et al., 2016; Tumber et al., 2014). In two studies (Scheerer et al., 2016; Tumber et al., 2014), subjects performed a rapid serial visual processing (RSVP) task while receiving pitch-shift alterations in their online auditory feedback during speech vowel vocalizations, while a different study (Liu et al., 2015) utilized a number of flash lights as the attention-dividing visual stimuli during the same speech AAF tasks. Despite existing differences in the implementation of divided-attention experimental paradigms (e.g., types and ranging complexity of visual stimuli), results of these studies have consistently reported a reduced magnitude of compensatory responses to pitch-shifted auditory feedback alterations when subjects were instructed to pay attention to competing visual cues (e.g., letters or flashing lights) during steady speech vowel production (Liu et al., 2015; Scheerer et al., 2016; Tumber et al., 2014).

In addition, Liu et al. (2018) conducted a follow-up study in which the load of attention-dividing stimuli was systematically manipulated via changing the presentation rate of irrelevant flash light visual cues during the same speech AAF tasks. Results of this study showed that subjects produced larger speech compensatory responses to pitch-shifted auditory feedback during the high vs. low attentional-load condition. This latter finding

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was rather paradoxical in that one would anticipate smaller speech compensations in the high-load condition where there is increased demand on attentional resources for processing the irrelevant visual cue stimuli. As discussed by Liu and colleagues (Liu et al., 2018), one possible account of the observed effect is the potential interaction between working memory, divided attention, and sensorimotor mechanisms of speech auditory feedback control. In this context, it has been proposed that the increased demand for the allocation of resources to working memory for storing sensory stimuli during a more complex divided-attention task may increase the overall neural capacity for the sensorimotor mechanisms implicated in speech error detection and motor control (Liu et al., 2018). While our understanding about the nature of interplay between such mechanisms remain mostly speculative, findings of these previous studies corroborate the notion that attentional instructions influence the underlying sensorimotor mechanisms involved in speech auditory feedback control.

The contribution to the understanding of the potential influence of higher-level cognitive mechanisms (e.g., attention) on speech function has important implications for developing new theoretical perspectives and advancing clinical approaches for targeted diagnosis and treatment of communication disabilities in individuals with neurological disorders. In the present study, we aimed to address this need by designing an experimental paradigm to probe the effects of attentional instructions on the behavioral and neural mechanisms of speech auditory feedback control. This paradigm differed from the ones used in previous studies in that it involved instructing subjects to pay attention to relevant pitch-shift alterations in their real-time auditory feedback within the single-modality auditory system instead of dividing attention between relevant and irrelevant stimuli in the auditory and visual modalities, respectively. Therefore, our paradigm provided a unified framework for examining the effects of attentional instructions on speech auditory feedback control while ruling out the inherent variability associated with the presentation of irrelevant visual cues during speech production tasks. This approach was motivated by evidence from previous studies showing that selective attention improves speech perception under normal and adverse listening conditions via enhancing the ability to track the fine acoustic structures of presented auditory stimuli (Lambez et al., 2020; Lesenfants and Francart, 2020; Rimmele et al., 2015; Saltzman et al., 2021; Tóth et al., 2020). In addition, it has been demonstrated that such improvements in speech perception abilities are also present in clinical populations such as those with autism spectrum and developmental language disorders (Emmons et al., 2022; Shafer et al., 2007; Soskey et al., 2017). Moreover, other studies have shown that the external focus of attention on acoustic goals can facilitate efficient and automatic control of speech movements (Lisman and Sadagopan, 2013), and active attention to auditory stimuli elicits stronger neural activities as indexed by the increased amplitude of event-related potentials (ERPs) (Hink and Hillyard, 1976; Stevens et al., 2006), as well as increased hemodynamic responses in cortical auditory networks (Ahveninen et al., 2006; Johnson and Zatorre, 2006; Sabri et al., 2008).

1.3. Study goals and hypotheses

In the present study, we used an experimental paradigm in which subjects in one group were explicitly instructed to pay attention to and detect pitch-shift alterations in their online speech auditory feedback while no such instructions were provided to the other

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(i.e., control) group. This experimental design allowed us to study the effects of attentional instructions on the behavioral and neural correlates of speech error correction and motor control in response to pitch-shifted auditory feedback alterations. As evidenced by previous studies, we hypothesized that cortical neural responses to AAF, as indexed by ERPs, would exhibit stronger modulations in the group that received attentional instructions compared with the no-instruction group (Getzmann et al., 2016; Hink and Hillyard, 1976; Stevens et al., 2006). In addition, we hypothesized that there would be a direct relationship between the ERP and behavioral compensation responses to AAF with stronger associations for the attentional-instruction group. This hypothesis was motivated by the evidence that ERP components reflect how strongly the sensorimotor system responds to auditory feedback errors during speech production (Houde and Chang, 2015). Moreover, as indicated by previous data showing that the ability to correct for speech errors is diminished during divided attention (Liu et al., 2015, 2018; Scheerer et al., 2016; Tumber et al., 2014), we hypothesized that attentional instructions would result in modulation of speech AAF compensation behavior. Finally, we hypothesized to observe a stronger modulation of neural activities in sensorimotor brain networks implicated in speech auditory control in the attentional-instruction compared with no-instruction group. This latter hypothesis was supported by previous findings showing that attended stimuli elicit stronger brain activity in cortical auditory regions compared with non-attended tasks (Ahveninen et al., 2006; Johnson and Zatorre, 2006; Sabri et al., 2008).

2. Materials and methods

2.1. Subjects

Two independent samples of 21 (14 females; age range: 52–74 yrs; mean age: 62 yrs) and 23 subjects (17 females; age range: 55–87 yrs; mean age: 64 yrs) were recruited and randomly assigned to the attentional-instruction and no-instruction groups, respectively. Subjects in both groups had no history of speech, language, hearing, or neurological disorders and were recruited from the greater Columbia, SC area through flyers, social media, and word-of-mouth. All subjects passed a binaural hearing screening at 40 dB threshold for frequencies at 250, 500, 1000, and 2000 Hz. The choice of study sample size was motivated by previously published data in a similar group indicating adequate statistical power at $1\text{-}\beta > 0.80$ while controlling for $\alpha < 0.05$ for examining the speech compensation magnitude (Cohen's d = 1.82, $t_{crit}(20) = 2.086$, $p < 0.05$, $1\text{-}\beta = 1.0$) as well as neurophysiological ERP responses to auditory feedback pitch-shift stimuli (Cohen's d = 0.89, $t_{crit}(20) = 2.086$, $p < 0.05$, $1\text{-}\beta = 0.97$) (Behroozmand et al., 2022). Informed consent was obtained from all subjects, and they were compensated for their participation time. Research protocol was approved by the University of South Carolina Institutional Review Board.

2.2. Experimental paradigm

The experiments were conducted in a sound-attenuated booth in which subjects' electroencephalography (EEG) and speech signals were recorded while performing speech production tasks under the AAF paradigm. For the AAF paradigm, subjects were instructed to follow the picture of a human face visual cue on the computer screen and produce steady vocalizations of the speech vowel sound /a/ for approximately 2–3 seconds at

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their conversational pitch and loudness levels (Fig. 1). Following each vocalization trial, a listening trial was implemented and cued by presenting the picture of a human ear on the screen while subjects silently listened to the playback of their own vocalization recorded in the previous trial. During both vocalization and listening trials, a brief (200 ms) pitch-shift stimulus at ±100 cents (100 cents = 1 semitone) magnitude was delivered to randomly alter the auditory feedback in upward or downward directions at randomized times between 750 and 1250 ms after the onset of vocalizations. In music theory, a semitone represents the smallest perceptible difference in pitch between two adjacent notes in the Western musical scale. In addition, control trials with no pitch-shift stimuli were randomly interleaved between AAF trials during both vocalization and listening tasks. This approach was implemented to prevent subjects from anticipating altered and unaltered trials during the experiment. Subjects in the attentional-instruction group were explicitly instructed to focus their attention on their auditory feedback and press a green button (Yes) if they detected an upward or downward pitch shift and press a red button (No) to indicate the absence of pitch-shift stimuli (i.e., control). Subjects performed the button press task approximately 4–5 seconds after the completion of each vocalization or listening trial, separately. For the no-instruction group, subjects were only asked to follow the visual cues to perform speech vowel vocalization and listening without any additional attentional instructions for pitch-shift detection and button press tasks. Training was given to both groups prior to the start of data collection to ensure that subjects understood the tasks and were able to perform them correctly. A total of 210 vocalization and 210 listening trials were collected for each subject, with each task comprising randomized presentations of upward and downward pitch-shift stimuli, as well as no-pitch-shift (control) trials. This resulted in 70 trials per stimulus type for both vocalization and listening tasks, separately. As motivated by our specific hypotheses, in the present study, we only analyzed the data for vocalization trials to investigate the effect of attentional instructions on the neural and behavioral compensatory responses to auditory feedback alterations during speech vowel sound production.

2.3. Speech data acquisition and analysis

The subjects' speech signals were recorded by a head-mount AKG condenser microphone (model C520), amplified by a Motu Ultralite-MK3, and recorded on a laboratory computer at 44.1 kHz sampling rate. A custom-design program in Max/Msp (Cycling 74, v.5.0) was used to control an Eventide Eclipse Harmonizer for delivering randomized upward (+100 cents) and downward (−100 cents) pitch-shift stimuli to speech vowel vocalizations. The online auditory feedback signal was delivered to subjects' ears via Etymotic insert earphones (models ER1–14 A). The Max/Msp program controlled various aspects of visual cues and pitch-shifting stimuli (e.g., duration, direction, amplitude, and timing), and generated transistor-transistor-logic (TTL) pulses to mark the onset of each event during speaking and button press tasks for synchronization with concurrently recorded EEG signals. The behavioral measures of subjects' speech compensatory response to upward and downward pitch-shift stimuli were calculated using the following procedures: First, the Praat software (Boersma and Weenink, 2009) was used to extract the pitch frequency contours of the recorded speech signals, and data were then exported to MATLAB to segment the extracted contours into epochs from −100 ms before to 500 ms after the onset of AAF pitch-shift stimuli. For each trial, the magnitude of speech compensation responses was calculated

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normalized to pre-stimulus baseline by converting pitch contours from Hertz to Cent scales using the following formula:

$$Speech\ Compensation\ Magnitude[Cent] = 1200 * log_2\left(\frac{F}{F_{Baseline}}\right)$$

Here, F is the post-stimulus pitch frequency and $F_{Baseline}$ is the baseline pitch frequency from −100 to 0 ms pre-stimulus. Response trials with speech compensation magnitudes exceeding ±200 cents were excluded from the analysis. The choice of this exclusionary threshold was based on the inspection of individual trials to remove outlier or artifactual responses (e.g., due to voluntary vocal pitch changes to feedback alterations) falling approximately ±2 standard deviations away from the mean. Speech responses to feedback alterations were combined for both opposing and following responses to pitch-shift stimuli (not analyzed separately). For each individual subject, the overall magnitude of speech compensation was calculated by averaging pitch contours across at least 100 trials combined across upward and downward pitch shift stimuli (minimum 50 trials for each stimulus direction). The sign of vocal responses to upward pitch shifts was flipped over, ensuring that the compensation magnitudes for both stimulus directions shared the same polarity. The grand-average speech compensation responses were calculated by averaging individual responses for subjects in the attentional-instruction vs. no-instruction groups, separately. The magnitude of speech compensation was measured by calculating the mean within a time window at 150–400 ms after the onset of pitch-shift stimuli. The choice of this time window was motivated by visual inspection of behavioral data to capture the peak of speech compensation responses for the two groups consistent with methods used in previous studies (Behroozmand et al., 2018, 2022; Sangtian et al., 2021).

To assess the impact of attentional instructions on speech auditory feedback error detection, the accuracy rates of button press responses to pitch-shifted trials were calculated for each subject in the corresponding group according to the following formula:

$$Accuracy\ Rate[\%] = \frac{Number\ of\ correct\ error\ detection\ trials}{Total\ number\ of\ response\ trials} \times 100$$

Here, the number of correct error detection trials indicates Yes button press responses to pitch shifts in the auditory feedback relative to the total number of Yes and No responses for all trials.

2.4. EEG data acquisition and analysis

EEG signals were concurrently recorded during the speech AAF tasks in both groups from 64 scalp electrodes following a standard 10–10 montage using the BrainVision active electrode system (Brain Products GmbH, Germany). The PyCorder software integrated with the BrainVision actiCHamp amplifier was used to record EEG signals at 1 kHz sampling rate after applying a low-pass anti-aliasing filter with 200 Hz cut-off frequency. Electrode impedances were kept below 5 kΩ for all channels. The EEGLAB toolbox (Delorme and Makeig, 2004) was used to interpolate artifactual data from noisy channels affected by high impedance or defected electrodes, and to filter the EEG signals at frequency ranges

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between 1 and 30 Hz (−24 dB/oct). The Independent Component Analysis (ICA) was applied to remove eye movement, blinks, muscle, and line noise artifacts. Following these pre-processing steps, the EEG signals were segmented into epochs ranging from −200 ms before to 500 ms after the onset of the pitch-shift stimuli, and baseline correction was performed by removing the mean amplitude of the pre-stimulus neural activities in a time window from −200 to 0 ms on a trial-by-trial basis for each channel. The choice of the baseline was based on visual inspection to identify the pre-stimulus time window within which the evoked neural activities were relatively stable. Individual event-related potentials (ERPs) were calculated by averaging the pre-processed EEG data for each subject across at least 100 trials combined across upward and downward pitch shifts (minimum 50 trials for each stimulus direction). In our analysis, we primarily focused on examining the effect of attentional instructions on the underlying mechanisms of speech feedback control irrespective of directional effects associated with pitch-shift stimuli. Therefore, for both the magnitude of speech compensation and amplitude of ERPs, responses to upward and downward pitch-shift directions were combined for statistical analysis. This allowed us to maintain statistical power by minimizing the number of factors and focusing on those with higher priorities relevant to our study questions and hypotheses. In addition, this approach was critical to provide a holistic account of the observed effects and allowed us to consistently (and concurrently) examine the neural correlates of behavioral effects using ERP by averaging data over a larger number of trials (i.e., minimum 100 trials per condition for each group, important for enhancing signal-to-noise ratio for ERP analysis). The grand-average measures of neural activity in response to pitch-shift stimuli were calculated by averaging individual ERP responses for subjects in the attentional-instruction *vs.* no-instruction group, separately. However, since ERPs are only elicited in response to a stimulus, we did not measure neural responses to no-alteration trials where there were no pitch shifts in the auditory feedback during speech vowel vocalizations.

The amplitude of the N1 and P2 components was measured at 130–170 ms and 230–350 ms in response to pitch-shift AAF stimuli, respectively. The choice of the ERP components was motivated by evidence supporting the notion that the N1 and P2 responses reflect how strongly the speech system responds to auditory prediction errors during the AAF experimental paradigm (Houde and Chang, 2015). The amplitude of N1 and P2 ERP responses were calculated by averaging neural activities across 22 fronto-central electrodes in each subject (FP1, FP2, AF7, AF3, AFz, AF4, AF8, F3, F1, Fz, F2, F4, FC3, FC1, FCz, FC2, FC4, C3, C1, Cz, C2, and C4). The choice of electrodes was based on visual inspection of topographical maps to identify areas showing the most prominent distribution of neural activities for the two ERP components.

2.5. Statistical analysis of speech compensation and ERP responses

We used an analysis of covariance (ANCOVA) model in SPSS (v.28) to examine the effect of attentional instructions on speech compensation magnitude, and to determine the association between speech compensation magnitude and neural activity as indexed by the ERP components. In this ANCOVA model, we examined the effect of group (attentional-instruction *vs.* no-instruction) as the independent variable on the magnitude of vocal compensation as the dependent variable while controlling for the variability accounted for

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by the N1 and P2 ERP components as the covariates. Effect size was estimated using partial etasquared (η_p^2) while controlling for Type-I error at $\alpha < 0.05$. The relationship between the magnitude of speech compensation *vs.* group and ERP responses was examined using a general linear regression model using the following formula:

$$Compensation_i \sim \beta_0 + \beta_1 * Group_i + \beta_2 * ERP_i + \varepsilon_i$$

Here, $Compensation_i$ is the magnitude of speech compensation as the dependent variable, and $Group_i$ and ERP_i are the experimental groups (attentional-instruction *vs.* no-instruction) and the amplitude of the ERP neural response components as independent variables, respectively. In addition, ε_i represents the residuals between the predicted value by the model and the actual value for the ith subject in each group. β_0, β_1, β_2 are the estimated coefficients of the corresponding predictors. Based on the results general linear regression model, the coefficient of determination (R-squared) was utilized as a measure to assess the strength of relationship between speech compensation magnitude and ERP neural activity responses in the attentional-instruction and no-instruction groups. Normality and homogeneity of variance of residuals assumptions were examined using the Shapiro-Wilk and Breusch-Pagan tests, respectively. Violation of the homogeneity assumption was detected for speech compensation responses, and therefore, a square-root transformation was applied to the dependent variable. The square-root transformation was a special case of the Box-Cox transformation with $\lambda = 0.5$ (Box and Cox, 1964), which did not change the monotonicity of data. Then the Shapiro-Wilk and Breusch-Pagan tests were performed again to confirm that the fits did not deviate from model assumptions. These procedures were conducted to minimize the potential impact of outliers and to ensure the overall robustness of the statistical model and the validity of outcome results. In addition, a two-sample independent *t*-test analysis was conducted to examine the effect of attentional instructions on neural activities as reflected by the amplitude of N1 and P2 ERP components. Effect size was estimated using Cohen's d measure while controlling for Type-I error at $\alpha < 0.05$.

2.6. ERP source localization

Modulation of cortical neural activity sources for altered auditory feedback in different groups (attentional-instruction *vs.* no-instruction) was determined based on cortical distribution differences in ERP current source densities extracted using the sLORETA algorithm. sLORETA solves the inverse problem of EEG source localization by generating a reference-free and polarity-independent single linear solution based on the linearly weighted sum of all scalp electrode potentials in the montage (Pascual-Marqui, 2002). For consistency, we extracted the profiles of the most salient neural activities averaged within the same time window at 130–170 ms and 230–350 ms to estimate the cortical sources of N1 and P2 ERP components for subjects in the attentional-instruction and no-instruction groups, respectively. Previous studies have validated the accuracy of the sLORETA source localization algorithm with the fMRI (Mulert et al., 2004), PET (Pizzagalli et al., 2004) and intracranial ECoG recordings (Zumsteg et al., 2006). For each subject, voxel-based sLORETA images were generated using realistic normalized head phantoms and Montreal Neurological Institute's standardized brain templates (MNI152 templates), and the 3D

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solution space was restricted to cortical grey matter. The MNI templates are used for spatial normalization and registration of neuroimaging data. They provide a common coordinate system, enabling consistent comparisons and facilitating collaboration with other studies in neuroscience. Voxel-by-voxel comparisons of ERP current source density distributions between attentional-instruction and no-instruction groups were performed, and results were corrected for multiple comparisons using non-parametric voxel-based statistical tests with 5000 random permutations.

3. Results

3.1. ERP neural responses

The overlaid profiles of grand-average ERP responses for the attentional-instruction and no-instruction groups performing the speech AAF task are shown in Fig. 2A. As shown in the topographical maps of Fig. 2B, the N1 and P2 components in response to AAF pitch-shift stimuli were elicited as neural activities with negative and positive polarities with predominant distributions over the fronto-central electrodes, respectively. A two-tailed independent-sample *t*-test conducted on the amplitude of P2 component at 230–350 ms averaged over the frontocentral electrodes revealed a significant effect of group with larger responses for the attentional-instruction compared with no-instruction ($t_{(42)} = 2.187$, $p < 0.05$, Cohen's d = 0.66) (Fig. 2C). However, no such effect of condition was revealed for the N1 ERP component ($t_{(42)} = 0.854$, $p = 0.398$, Cohen's d = 0.258).

3.2. Speech compensation responses

The overlaid profiles of grand-average speech compensation responses for the attentional-instruction and no-instruction groups performing the AAF task are shown in Fig. 3A. For subjects in the attentional-instruction group, the average speech error detection accuracy rate was 75% (SD: 35%) with the majority (15 out of 21) exhibiting high accuracy rates (>80%) while 3 subjects showed moderate (40–80%) and 3 subjects showed low accuracy rates (<40%). The ANCOVA model revealed a significant effect of group (attentional-instruction *vs.* no-instruction) on speech compensation responses between 150 and 400 ms after the onset of pitch-shift stimuli ($F_{(1,39)} = 21.1$, $p < 0.001$, $\eta_p^2 = 0.351$) (Fig. 3B). We found that this effect was associated with smaller magnitudes of speech compensation responses for the attentional-instruction compared with the no-instruction group while controlling for the amplitude of the ERP components. No association was found between the amplitude of the N1 ERP component and the magnitude of speech compensation responses while controlling for the group variable in the model ($F_{(1,39)} = 0.052$, $p = 0.82$, $\eta_p^2 = 0.001$) (Fig. 3C). However, we found a significant association between the amplitude of the P2 ERP component and the magnitude of speech compensation responses ($F_{(1,39)} = 12.90$, $p < 0.001$, $\eta_p^2 = 0.249$) (Fig. 3D). The linear regression model revealed a positive association between the amplitude of the P2 ERP component and the magnitude of speech compensation responses in both groups and yielded an R-squared value of 0.41, indicating that experimental group and P2 ERPs accounted for 41% of variability in speech compensations.

To further evaluate the relationship between the P2 component and speech responses while controlling for between-subject variability, two additional models were employed

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to replicate the regression analyses by splitting subjects' data into upper and lower halves based on the magnitude of their compensations: 1) The analysis of upper-half responses revealed a significant positive linear association between the amplitude of P2 ERP components and speech compensation magnitudes ($p = 0.0001$), indicating consistent results with our main model findings. Notably, as shown earlier, the magnitude of speech compensation responses in the attentional-instruction group was significantly smaller than those in the no-instruction group ($p < 0.0001$). 2) The analysis of lower-half responses showed a marginally significant positive linear association between the amplitude of P2 ERP components and speech compensation magnitudes ($p = 0.0583$). Given the fact that splitting the data into upper and lower halves would inevitably reduce statistical power, it is reasonable to consider a less stringent statistical criterion and determine this latter effect as an indication of a meaningful relationship between the neural and behavioral correlates of speech auditory feedback control. In addition, as shown earlier, the magnitude of speech compensation responses in the attentional-instruction group was significantly smaller than those in the no-instruction group ($p = 0.0146$).

3.3. ERP source localization

Results of sLORETA analysis for P2 ERP responses to pitch-shift AAF revealed significantly stronger cortical neural activity in the attentional-instruction compared with no-instruction groups in areas within the cortical auditory networks including the superior temporal gyrus (STG) and transverse temporal gyrus (TTG), as well as motor (i.e., pre-central gyrus [PrCG]), somatosensory (i.e., post-central gyrus [PoCG]), and insular cortex ($t > +3.50$, $p < 0.05$). In Table 1, the Montreal Neurological Institute (MNI) coordinates for voxel peaks within anatomically distinct clusters of significant between-group modulation of P2 neural activity (attentional-instruction > no-instruction) and their corresponding Brodmann's areas (BA) are reported. The sources of P2 neural activity are visually demonstrated using a color-coded heat map in Fig. 4, where areas with dark red and yellow colors show significant between-group effects ($t > +3.50$, $p < 0.05$).

4. Discussion

In this study, we used an AAF experimental paradigm to examine the effects of attentional instructions on the underlying behavioral and neural mechanisms of speech auditory feedback control in response to pitch-shift stimuli. Our data revealed stronger P2 ERP neural responses to speech auditory feedback alterations in the attentional-instruction as compared with no-instruction group. This finding is in support of our hypothesis and aligns with previous reports on the modulatory effects of attentional instructions, where attended stimuli were shown to elicit stronger auditory neural activities during speech production and comprehension tasks (Getzmann et al., 2016; Hink and Hillyard, 1976; Rimmele et al., 2011; Stevens et al., 2006). Moreover, previous evidence has indicated that the amplitude of P2 ERPs is systematically decreased under conditions in which attentional instructions are implemented to increasingly divided neural resources towards concurrent and yet irrelevant AAF stimuli during speech production (Liu et al., 2018).

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For tasks involving speech feedback control, the P2 ERP component has been proposed to represent higher-level sensorimotor neural processes implicated in auditory error detection and correction (Houde and Chang, 2015). Consistent with findings from a previous study (Scheerer et al., 2013), our data support this proposal by showing a significant positive relationship between the amplitude of P2 ERP neural activities and the magnitude of speech compensation in response to pitch-shifted auditory feedback in both the attentional-instruction and no-instruction groups. In this context, the P2 component has been suggested to receive contributions from higher-level cortical auditory-motor networks within the posterior peri-Sylvian regions (Crowley and Colrain, 2004; Hari et al., 1990), including the posterior superior temporal sulcus (pSTS) and area Spt at the temporo-parietal boundary, which serve as a potential neural interface for speech sensorimotor integration (Hickok et al., 2009). Converging evidence from several studies has supported this view by showing that the top-down effects associated with speaking-induced modulation of neural activities in response to AAF stimuli are predominantly observed in components that emerge at longer latencies overlapping with P2 ERPs (Behroozmand et al., 2009; Chang et al., 2013; Kort et al., 2014). Accordingly, such integrative processes are hypothesized to play a key role in comparing the efference-based sensory representations of the intended speech output with their actual auditory feedback for sensory prediction error processing (Houde and Chang, 2015). In addition, our data suggests that P2 responses may also represent the neural state of the internal model that translates auditory prediction errors into corrective motor commands for speech compensation. However, while this is a plausible interpretation, the relative contribution of the underlying sensory error detection and motor correction mechanisms to P2 ERPs are not clearly understood. This is partly due to the fact that prediction errors driving speech compensations could not be directly quantified in the present study. Therefore, the above arguments should be considered with caution while interpreting the observed effects in the context of our hypotheses.

Based on the observed between-group effects, we suggest that our data provide preliminary evidence that attentional instructions can improve the mechanisms of speech error detection and sensorimotor control. This notion is supported by findings of the present study showing that the fronto-central P2 ERPs in response to AAF stimuli were elicited with significantly stronger amplitudes in the group that received corresponding attentional instructions compared with their no-instruction counterparts. This effect reflects a complex interplay between neural processes implicated in the allocation of attentional resources and those involved in speech auditory feedback control. Although a mechanistic account of cross-modal interactions between these systems remains elusive, we argue that attentional instructions may consequently induce top-down modulatory effects that selectively enhance auditory neural sensitivity for detecting mismatches (i.e., errors) between the predicted and actual sensory representations of intended speech. Supporting evidence for this notion is provided by recent studies in humans showing that top-down activation of frontal attentional networks can selectively modulate the tuning properties of neurons in the auditory cortex (Chou and Sen, 2021; Lesenfants and Francart, 2020; Mesgarani and Chang, 2012; Wikman et al., 2021). This effect may consequently result in the refinement of sensorimotor transformations, thereby aiding the optimization of corrective motor behavior in response to speech auditory feedback error signals.

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Our findings revealed that while the P2 component of neural responses to AAF was enhanced in the attentional-instruction group, the magnitude of speech compensation was significantly reduced in these subjects compared with the no-instruction group. This rather counter-intuitive finding provides new insights into the potential effects of attentional instructions on speech production and motor control. In the following sections, we provide several interpretations that may potentially account for the observed effects in our study. Firstly, we propose a model that incorporates attentional instructions as a gain-modulating factor in speech sensorimotor integration mechanisms (Fig. 5). According to this model, the observed increases in P2 neural activities and localization of corresponding sources to the right auditory (i.e., STG and TTS) and somatosensory (i.e., post-central gyrus) cortical networks provides indication for enhanced sensory processing of speech feedback errors in the attentional-instruction compared with no-instruction group. This effect can in part be accounted for by a greater involvement of higher-level cognitive mechanisms (e.g., attention) in the former group, leading to enhanced neural processing of auditory and somatosensory feedback changes arising from pitch shifts and subsequent compensatory movement of laryngeal muscles for speech motor control, respectively (Chou and Sen, 2021; Lesenfants and Francart, 2020; Mesgarani and Chang, 2012; Wikman et al., 2021). Consequently, increased neural activities within the right post-central gyrus in the attentional-instruction group may account for a larger error signal represented in the motor system. This effect may in turn result in an adaptive modulation of compensation gain for generating corrective motor commands in response to feedback errors. As supported by our data, we suggest that the compensation gain follows a Gaussian distribution function (see yellow curve in Fig. 5), which assigns smaller gains to larger deviations (i.e. errors) from the intended speech output, as determined by the efference copies of motor commands. Converging evidence from previous studies support this notion by showing that speakers generate larger compensatory responses for smaller changes in their auditory feedback (Behroozmand et al., 2009; Liu and Larson, 2008). For example, Behroozmand et al.'s study (Behroozmand et al., 2009) reported an increase in auditory responsiveness to the 100 cents compared to 500 cents pitch-shift stimuli. In addition, speaking-induced modulation of neural activities are shown to be largest when feedback deviations are small and closely match the intended output (Behroozmand et al., 2009). In this context, the adaptive modulation of compensation gain has important functional implications for allowing speakers to optimally correct for self-generated feedback errors while avoiding destabilizing behavior by suppressing corrective motor responses to larger deviations induced by external sources (e.g., environmental sounds and those produced by other speakers). In this context, we suggest that our data corroborate the notion that attentional instructions may lead to increased sensitivity to feedback changes, resulting in neural interpretations of a larger error signal in response to the same level of pitch-shift alterations during speech production. This effect would then result in assigning a lower compensation gain, which would lead to smaller corrective speech motor response to AAF stimuli.

Although our proposed model provides a possible explanation of the observed behavioral and neural consequences of attentional instructions, it is noteworthy that its principles remain mostly speculative and will require further validation using evidence from future studies. This is mainly because of several limitations outlined in the following section.

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Firstly, while attentional instructions led to increased neural responses to AAF stimuli, our data did not provide direct evidence for an increase in the size of perceived feedback errors. This is partly because although P2 ERPs were positively correlated with speech compensation magnitudes, the underlying error signals driving such behaviors are not directly observable. In addition, the relation between feedback errors and compensation magnitudes is greatly affected by the properties (e.g., slope) of the gain function. For example, with a steep-enough slope, compensatory responses to a large feedback error can be potentially smaller than those in response to a smaller error. This implies that there can be different combinations of error signals and gain functions leading to a similar speech compensatory response. Moreover, the complex interplay between attentional instructions and higher-level cognitive and sensorimotor mechanisms could not be examined in the present study. This is in part due to an inherent limitation in our experimental design, which did not provide a direct measure to determine the extent of attentional modulations between subjects in the two groups. Therefore, future studies are warranted to probe the influence of instructions on higher-level attentional and sensorimotor mechanisms during speech auditory feedback control. Some studies have also suggested that performing an active motor behavior may affect the neural state of brain networks involved in sensory processing (Reznik and Mukamel, 2019). While it is reasonable to assume that this may affect the neural processing of pitch shifts in the attentional-instruction group who performed an additional button press task, the temporal separation between the vocalization and listening tasks *vs.* the button press responses in these subjects would minimize potential confounds associated with motor-related cross-modal influences. Specifically, subjects in the attentional-instruction group registered their responses via a button press that was performed a few (i.e., 4–5) seconds after the vocalization and listening tasks were completed, which minimized the temporal overlap between motor planning and execution of upper limb (i.e., hand) movements *vs.* pitch-shift responses to speech vowel productions. In addition, the recruitment of predominantly older adult subjects in this study raises the question whether including a younger group would have yielded different results due to potentially normative attentional abilities in younger individuals (Kok, 2000; Madden, 2007; Sekuler et al., 2008). Also, it should be noted that three out of the twenty-one subjects had low accuracy rates in the button press task, possibly attributed to fatigue among the aging subjects. Furthermore, the present study utilized a between-group design by comparing data between two groups of subjects who performed the tasks under attentional-instruction and no-instruction conditions. Although, as compared to a within-group design, the choice of this experimental approach raises potential concerns regarding the inherent individual variability for the behavioral and neural responses to AAF stimuli, we aimed to mitigate such confounding effects by comparing data between two randomly assigned age- and gender-matched groups in our study. Further, we conducted assumption checks and applied appropriate transformations to ensure that the residuals were normally distributed and had equal variances during statistical analysis. Moreover, we performed supplementary analyses to account for between-subject variations by examining the linear association between the behavioral and neural responses while splitting the data into upper and lower halves based on the magnitude of speech compensatory responses. These additional analyses further validated our findings by showing a consistent pattern of significant positive linear association between the amplitude of P2 ERP components and the magnitude of speech compensations in the regression

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models. In addition, consistent with earlier results, we found that compensation magnitudes were significantly diminished in the attentional-instruction compared to no-instruction group. These additional analyses further validated our earlier findings and corroborated our interpretation of the role of attentional instructions in modulating the underlying mechanisms of speech auditory feedback control. Our data provide new insights into the potential implications of utilizing attentional instructions as a method for enhancing speech feedback error compensation mechanisms. However, despite such efforts, it remains a possibility that unaccounted individual variability may, at least partially, confound the results. Therefore, because of the lack of determinacies and potential limitations, caution should be considered while interpreting the observed effect in the context of the proposed model.

The dissociation between the neural and behavioral results in our study can also be discussed in the context of the dual-stream model of speech (Hickok & Poeppel, 2004, 2007). In this model, the dorsal stream provides an integrative sensorimotor interface to compare predicted and actual sensory representations for speech control while the ventral stream is primarily implicated in comprehension of incoming sensory information. Research on the dual-stream model of visual awareness has shown a dissociation between the ventral and dorsal streams where the ventral pathways are activated during conscious perception of sensory stimuli, while the dorsal pathways can process visually guided actions without conscious perception (Fang and He, 2005; Goodale and Westwood, 2004; Zhan et al., 2018). In addition, it has been shown that conscious awareness and increased confidence in speech feedback outcomes can lead to reduced reliance on dorsal streams for the integration of external auditory feedback (Subramaniam, 2021), thereby influencing (i.e., reducing) the extent to which motor compensation is generated in response to AAF stimuli. Thus, it is reasonable to assume that conscious processing of speech feedback errors is mediated by dissociated mechanisms in ventral and dorsal streams, leading to different behavioral consequences in the attentional-instruction *vs.* no-instruction group. In our study, we explicitly instructed one group to be attentive and report external pitch changes in their real-time speech auditory feedback, making them to be cognizant of and consciously perceive and detect AAF stimuli, while such instructions were not given to the other group. Therefore, subjects in the attentional-instruction group may have compromised sensorimotor processing of feedback error signals in the dorsal stream for conscious processing of those changes in the ventral stream networks. Such switching processes may limit neural processing capacity in dorsal pathways for error correction, providing an alternative explanation for the diminished magnitude of speech compensatory responses in subjects in the attentional-instruction group. Consequently, we posit that this may account for the observed reduction in compensatory response when subjects are instructed to pay attention to auditory feedback. Our data provide preliminary evidence for a potential link between attentional instructions, feedback error processing, and speech compensation; however, further research is warranted to fully understand the complex nature of relationship between these behavioral and neural mechanisms.

The analysis of ERP sources indicated significantly larger neural activities in a network comprising areas within the right insula, precentral gyrus (PrCG), postcentral gyrus (PoCG), superior temporal gyrus (STG), and transverse temporal gyrus (TTG) for subjects in the

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attentional-instruction as compared with the no-instruction group. This finding highlights the specialized role of attentional instructions in modulating right-hemispheric brain networks implicated in speech production and sensorimotor control. Converging evidence from previous studies has indicated the more predominant role of the right-hemispheric cortical auditory networks in pitch processing during speech production and singing (Riecker et al., 2000; Van Lancker and Sidtis, 1992; Zatorre et al., 1992). In addition, other studies have reported bilateral activation of cortical motor and insular networks during tasks involving vocal pitch modulation (Dichter et al., 2018; Perry et al., 1999). In this context, our findings regarding increased neural activity in right-hemispheric sensorimotor cortical regions (e.g., STG, TTG, PrCG, and PoCG) in the attentional-instruction group further highlight the significant role of these brain networks in vocal pitch regulation during speech production. In addition, the insular cortex has been proposed to be involved in ventral and dorsal stream processing of speech while receiving contribution from higher-level cognitive mechanisms during conscious processing of relevant sensorimotor stimuli (Fridriksson et al., 2016). Therefore, considering findings in our study, a plausible proposition is that heightened neural activity in the right insular cortex in the attentional-instruction group may reflect conscious perception of speech errors in the ventral stream networks. This notion is supported by the dual-stream model of speech, suggesting that while dorsal stream processing is predominantly mediated by left-lateralized brain networks, the ventral stream processing of speech is bilaterally represented in the human brain (Hickok et al., 2011; Hickok and Poeppel, 2004). In addition, increased neural activity in areas within the right STG and TTG in the attentional-instruction group further corroborates the notion of elevated conscious processing of speech auditory feedback errors in the ventral stream (Fridriksson et al., 2016; Hickok and Poeppel, 2015). Interestingly, we also found a concurrent pattern of neural activity increase in areas within the right PrCG, suggesting potential modulation of speech compensatory gain in cortical motor areas in response to auditory feedback errors in the attentional-instruction group. This effect was accompanied by elevated neural activities in somatosensory cortical areas within the right PoCG, potentially resulting from changes in proprioceptive feedback input arising from laryngeal muscles for vocal pitch control during speech vowel production. Together, these results highlight the effect of top-down attentional-instruction mechanisms in modulating neural activities in distributed right-hemispheric brain networks implicated in speech feedback error detection and correction.

There are several practical and clinical implications of our study findings that can inform future research and interventions for individuals with neurological conditions leading to speech disorders. Along this line, our study addresses a critical gap in previous literature by elucidating the effects of attentional instructions on the underlying behavioral and neural mechanisms of speech production and auditory feedback control. These findings suggest that instructions for maintaining attention to auditory feedback may be a useful tool for clinicians working with individuals with speech disorders, such as apraxia of speech or neurological disorders with language impairments, such as Parkinson’s disease and stroke (Behroozmand and Larson, 2011; Brendel et al., 2004; Maas et al., 2008; Patel et al., 2023). Clinical interventions involving attentional instructions may improve behavioral outcome measures while facilitating neural activities in key brain regions including frontal, temporal, and parietal cortices implicated in speech production and sensorimotor control. In this context,

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our findings highlight the potential application of attentional instructions for improving speech motor function in clinical populations suffering from neurological disorders.

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Data availability

Data will be made available on request.

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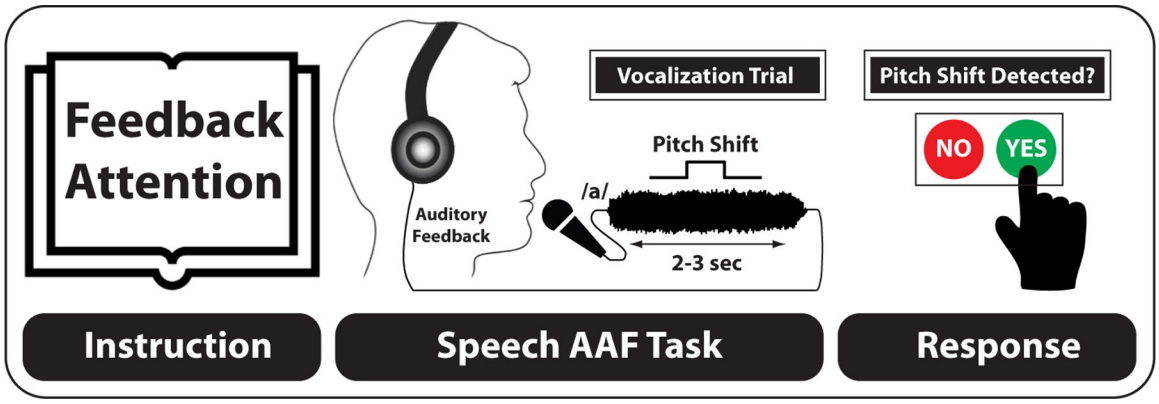

Fig. 1.

Altered auditory feedback (AAF) experimental paradigm for attentional instructions. Subjects produced steady vocalizations of the speech vowel sound/a/for approximately 2–3 seconds at their conversational pitch and loudness while receiving a brief (200 ms) pitch-shift stimulus at ±100 cents magnitude with randomized upward and downward direction 750–1250 ms following the onset of vocalizations. Subjects were explicitly instructed to pay attention to their speech auditory feedback and press a green button (Yes) to indicate the detection of pitch-shift stimuli and press a red button (No) for the absence of pitch shifts in control trials.

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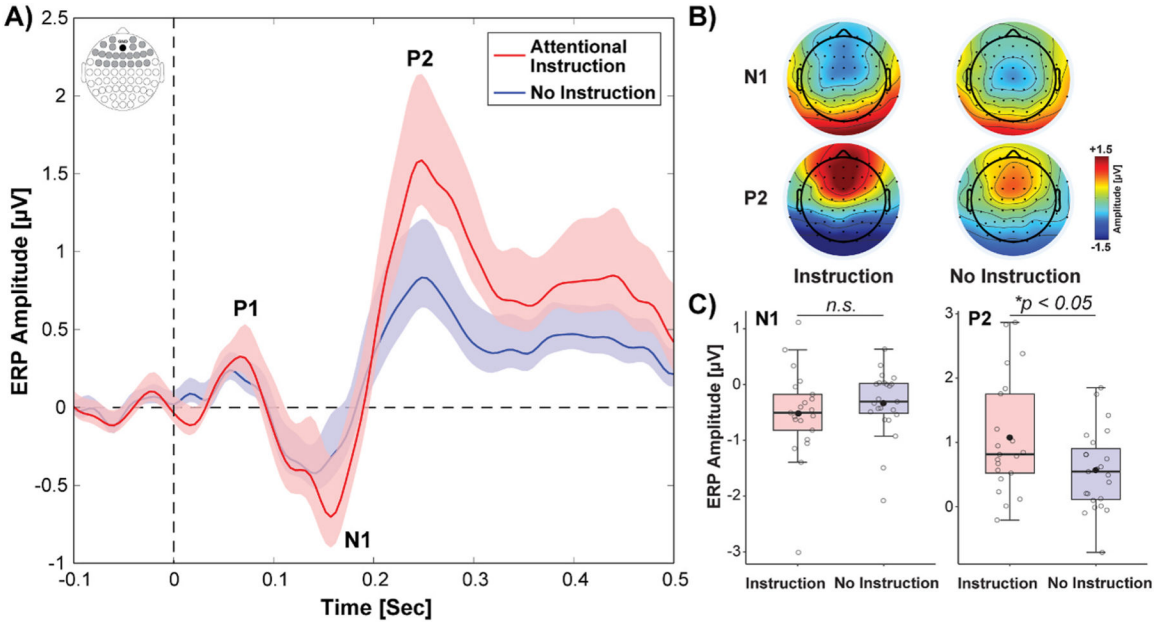

Fig. 2.

A) The overlaid temporal profiles of group-average ERP response to auditory feedback pitch-shift stimuli for 22 fronto-central electrodes overlaid across the attentional-instruction *vs.* no-instruction groups (the scalp location of fronto-central electrodes is shown on the top left corner). The time point at 0 ms indicates the onset of pitch-shift stimuli in the auditory feedback. **B)** The topographical maps of N1 and P2 components for the two groups. **C)** The boxplot representation of the N1 and P2 ERP amplitudes for the attentional-instruction vs. no-instruction groups. The box represents the interquartile range (IQR), the line inside the box is the median, the filled black circle inside the box is the mean, and the whiskers extend to the data range. The empty grey circles show individual subjects' data.

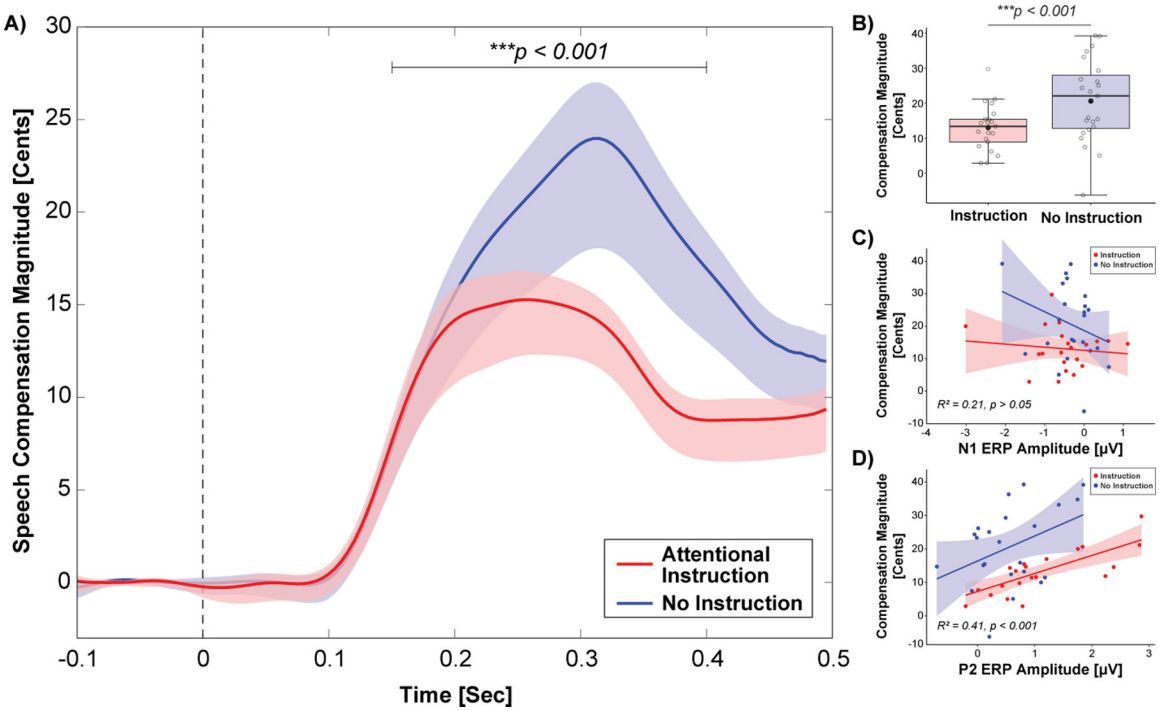

Fig. 3.

A) The temporal profiles of group-average speech compensation responses to pitch-shift stimuli overlaid for the attentional-instruction and no-instruction groups. Both upward and downward shifts are included, with responses to upward pitch shifts being flipped over to ensure consistent comparison. Timepoint 0 indicates the onset of pitch-shift stimuli. **B)** The compensation magnitude was measured within the time window between 150 and 400 ms after the onset of auditory feedback pitch-shift stimuli at time 0 ms. The boxplot shows the differences in the compensation magnitude in both groups. Outliers are retained in the figure. The box represents the interquartile range (IQR), the line inside the box is the median, the filled black circle inside the box is the mean, and the whiskers extend to the data range. The empty grey circles show individual subjects' data. **C)** The scatter plot for the relationship between the magnitude of speech compensation and N1 ERP amplitude. **D)** The scatter plot for the relationship between the magnitude of speech compensation and P2 ERP amplitude.

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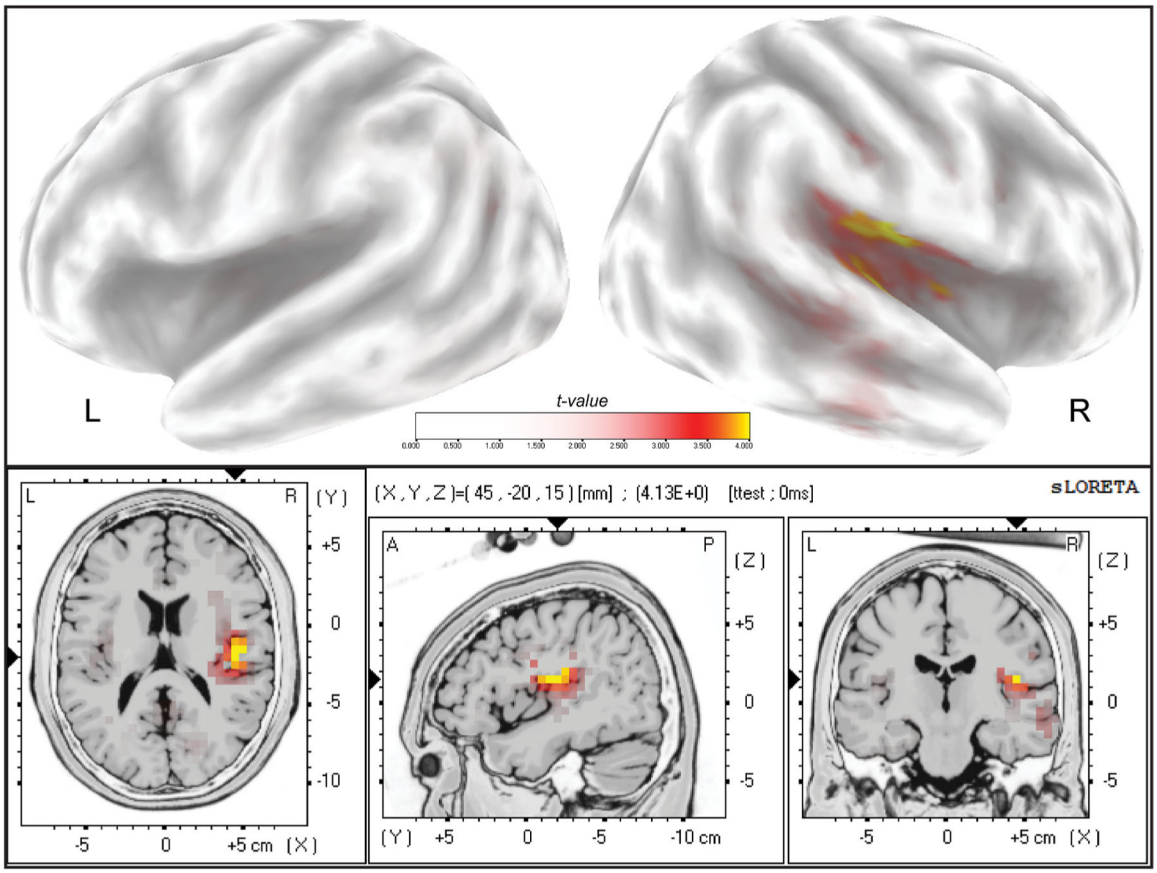

Fig. 4.

Neuroanatomical representation of voxel clusters with maximal between-group difference (attentional-instruction > no-instruction) of current source density for P2 ERP activations. In this figure, darker red and yellow colors show the cortical sources for areas with significant between-group increases in P2 neural activities for the attentional-instruction *vs.* no-instruction group ($t > +3.50$, $p < 0.05$).

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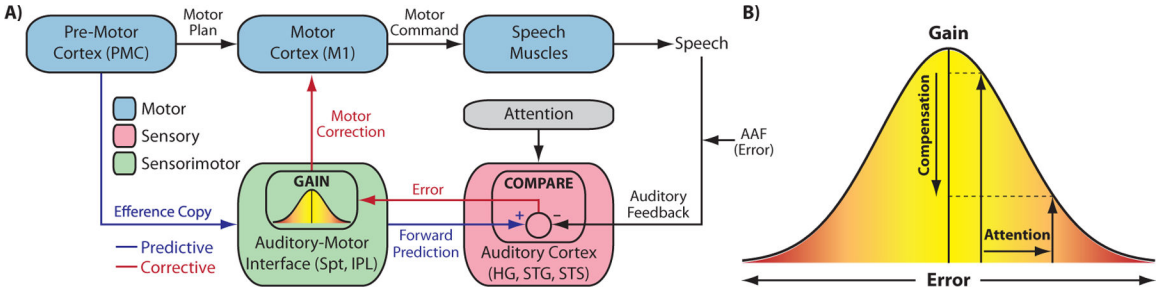

Fig. 5.

A) The attentional-instruction model of speech sensorimotor integration. During speech production, a motor plan is generated in the pre-motor cortex and transmitted to cortical motor areas. An efference copy of the motor plan is translated into a forward prediction code in the auditory cortex via the auditory-motor interface. The auditory cortex then compares the forward prediction with the speech output, generating an error signal in case of any mismatches. The auditory-motor interface translates the error signal from sensory to motor domain while assigning an adaptive compensation gain to correct for feedback changes. The compensation gain follows a Gaussian distribution function, which assigns larger gains to correct for smaller errors in the auditory feedback, and vice versa. Attentional instructions can increase neural sensitivity to feedback errors, which consequently leads to smaller speech compensations. **B)** The Gaussian function for adaptive speech compensation gain in response to auditory feedback errors. The x-axis represents the magnitude of error signals, and the y-axis shows the gain for speech motor compensation. For a given error signal, the compensation gain is determined by mapping the x-axis value onto the Gaussian curve on the y-axis (upward vertical arrows and dotted lines). Attentional instructions increase in the magnitude of error signals, leading to a reduction in speech compensation gain (downward vertical arrow). The Gaussian function demonstrates symmetrical properties, which similarly maps positive (i.e., predicted feedback > actual feedback) and negative (i.e., predicted feedback < actual feedback) auditory error signals onto corrective motor commands for speech compensation. AAF: altered auditory feedback; IPL: inferior parietal lobe; HG: Heschl's gyrus; STG: superior temporal lobe; STS: superior temporal sulcus.

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Table 1

The MNI coordinates and Brodmann's areas (BA) of voxel peaks within brain regions of significant between-group modulation of P2 ERP sources with stronger activity in the attentional-instruction *vs.* no-instruction group in response to speech altered auditory feedback. The critical t-value at $\alpha < 0.05$ was determined at +3.337.

Voxel (MNI)							
x	**y**	**z**	**t value**	**BA**	**Hemisphere**	**Lobe**	**Structure**
45	−20	15	+4.13	13	Right	Sub-lobar	Insula
50	−15	15	+4.07	43	Right	Parietal	Postcentral Gyrus
45	−25	15	+3.83	41	Right	Temporal	Insula
50	−25	20	+3.81	40	Right	Parietal	Postcentral Gyrus
50	−10	15	+3.76	43	Right	Frontal	Precentral Gyrus
45	−20	10	+3.54	13	Right	Temporal	Superior Temporal Gyrus
50	−20	10	+3.50	41	Right	Temporal	Transverse Temporal Gyrus